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ABSTRACT

Deepfake speech technology has sparked serious worries about the growing danger of audio manipulation, which could threaten national security as well as individual safety. The emergence of synthetic speech attacks has brought attention to the need for efficient detection techniques because current methods frequently lack accuracy and resilience, especially when it comes to complex GAN-generated false voices. By creating a comprehensive system that incorporates Generative Adversarial Networks (GANs) and Machine Learning (ML) algorithms for the creation and detection of false voices, this project tackles these issues.

The project uses a generator to produce realistic fake audio samples and a discriminator to assess the authenticity of the fake audio samples using GANs. Then, further data audio sets and this produced fake audio will be used for detection and training. For feature extraction, Mel-Frequency Cepstral Coefficients (MFCCs) are used, which help the classifier discover strong patterns and attain high detection accuracy. The procedure entails gathering and preprocessing audio data, training GANs to generate realistic audio, and then training the model with the generated audio. This study provides a strong defense against the increasing threat of audio deepfakes by demonstrating the potential of integrating GANs and ML approaches to improve the security and authenticity of audio communications.

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

1.1 Introduction

Finding the difference between real and false audios has become more challenging as deepfake technology advances. In order to overcome these difficulties, this study uses Generative Adversarial Networks (GANs) and Machine Learning (ML) models to generate and identify fake audio, with a special emphasis on voice-based audio. The two primary aspects are:

- A GAN for producing realistic-sounding synthetic audio samples.
- Audio features are used by a Machine learning algorithms to detect phony sounds.

Generative Adversarial Networks, or GANs, are neural networks that used for creating new voices.

Generative Adversarial Networks (GANs) are a powerful class of neural networks that are used for an unsupervised learning. A discriminator and a generator are the two neural networks that make up a GAN. To create synthetic data that is exact replicas of real data, they employ adversarial training.

By creating random noise samples, the Generator aims to deceive the Discriminator, which is responsible for correctly differentiating between created and real data.

As a result of this competitive interaction, both networks advance and realistic, high-quality samples are generated.

The widespread usage of GANs in text-to-speech synthesis, style transfer, and voice synthesis demonstrates how adaptable these artificial intelligence technologies are.

Architecture of GANs

A Generative Adversarial Network (GAN) is composed of two primary parts, which are the Generator and the Discriminator.

Generator Model

The generator model is an essential part of a Generative Adversarial Network (GAN) that generates new, precise data. Random noise is fed into the generator, which transforms it into intricate data samples. Usually, it's a deep neural network.

During training, many layers in a neural network capture the underlying distribution of the training data. As the generator is being trained, it uses backpropagation to fine-tune its parameters, adjusting its output to create samples that closely resemble real data.

What makes the generator effective is its capacity to produce diverse, high-quality samples that can deceive the discriminator.

Discriminator Model

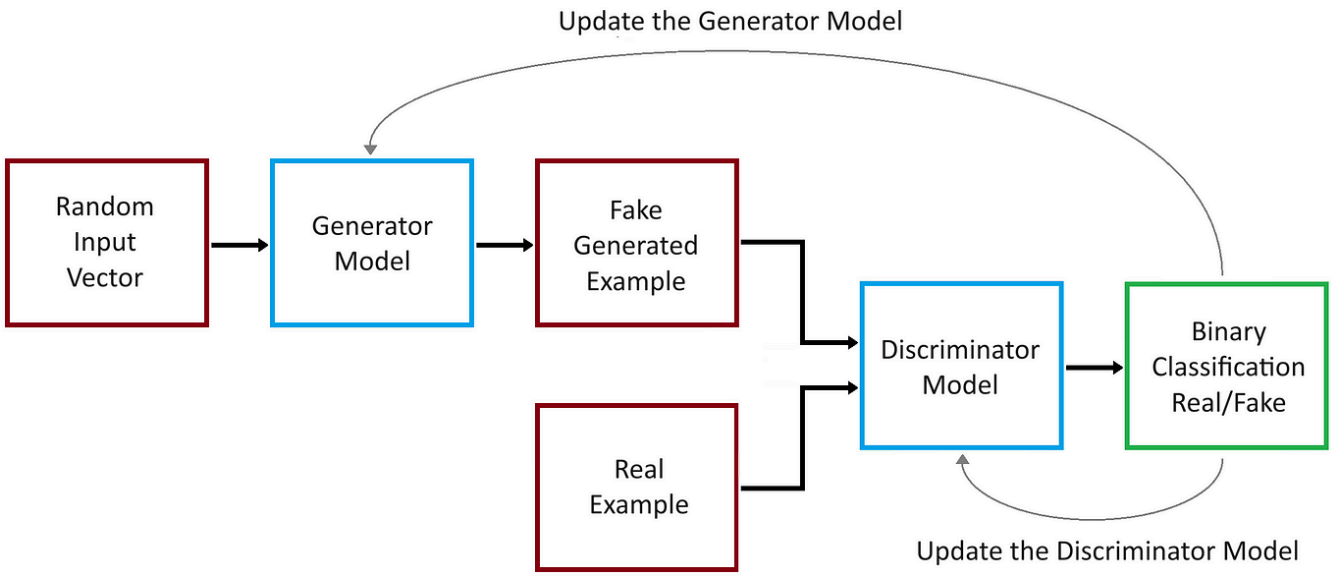
An artificial neural network called a discriminator model is used in Generative Adversarial Networks (GANs) to differentiate between generated and actual input.The discriminator acts as a binary classifier by evaluating incoming samples and assigning a probability of authenticity.

The discriminator gradually gains the ability to distinguish between real data from the dataset and fictitious samples produced by the generator. This enables it to gradually adjust its settings and raise its accuracy level.

In order to improve training stability and model performance, HiFi-GAN, a generative adversarial network for speech synthesis, is composed of two discriminators: multi-scale and multi-period discriminators, as well as one generator. Both discriminators and the generator are trained by adversarial means.
A fully convolutional neural network serves as the generator. It takes a mel-spectrogram as input and upsamples it using transposed convolutions until the output sequence length is equal to the original waveforms' temporal resolution. Multi-receptive field fusion (MRF) module comes after each transposed convolution.

The discriminator is a multi-period discriminator (MPD), which is made up of multiple sub-discriminators, each of which handles a subset of the input audio's periodic signals. Furthermore, MelGAN's multi-scale discriminator (MSD), which successively assesses audio samples at several levels, is employed to identify recurring patterns and long-term dependencies.

Fig 1.1 Architecture of GAN



1.2 Motivation

The necessity to deal with the rising threat posed by audio deepfake technology which may be misused for identity theft, disinformation, and privacy violations is what motivated this initiative. As deepfake speech synthesis has advanced, existing detection methods frequently find it difficult to distinguish fake voices, especially when confronted with sophisticated GAN-generated audio. By developing a dependable detection system that blends machine learning methods for robust identification with GANs for producing realistic fake samples, this study seeks to improve audio security. In the end, it strives to protect digital communications, stop the misuse of artificial audio in crucial contexts, identify and stop fraud, assist with international investigations, and provide entertainment, among other things.

1.3 Problem Statement

- As a means to improve the security and dependability of audio material, this project intends to create a system that uses Generative Adversarial Networks (GANs) to create realistic fake audio and Machine Learning (ML) techniques, notably a Random Forest classifier, to recognize such false audio.

- The objective is to create a comprehensive and adaptable Deepfake speech recognition model on GAN-generated audios that can be used in a variety of contexts, from the production of artistic material to real-world uses in industries like design and advertising.

1.4 Report Organisation

Chapter 1: Introduction

Chapter 2: Provides an overview of the Literature Survey.

Chapter 3: Gives an overview of technologies used.

Chapter 4: Gives details on the implementation.

Chapter 5: Deals with results of the proposed architecture.

Chapter 6: Conclusion and future work.

Chapter 7: Contains the references used.

CHAPTER 2

LITERATURE SURVEY

2. LITERATURE SURVEY

Recent advances in deep learning, generative adversarial networks (GANs), and audio signal processing techniques have significantly advanced the development of synthetic voice creation and the detection procedures that go along with it. By reviewing and evaluating three key publications, this literature review seeks to investigate noteworthy advances in this field: [1] "HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis," [2] "A Voice Cloning Method Based on the Improved HiFi-GAN Model," and [3] "Deepfake Audio Detection System," while also considering how these support and relate to the objectives of this project, which entails using HiFi-GAN to generate fake voices and machine learning (ML) algorithms to detect such fakes.

Using generative adversarial networks[6], the first paper under discussion, "HiFi-GAN," by Kong et al., suggests a unique architecture for synthesizing high-fidelity speech using mel-spectrograms. This study's main objective was to resolve the speed and audio quality trade-offs present in earlier models like WaveNet [5] and WaveGlow. HiFi-GAN presents a fully convolutional generator that uses transposed convolutions and multi-receptive field fusion (MRF) modules to convert mel-spectrograms into audio waveforms. The generator's ability to simultaneously record patterns of varying lengths is made possible by these MRF modules, which is essential for maintaining the periodic structure of natural speech.Two specialized discriminators, the Multi-Period Discriminator (MPD) and the Multi-Scale Discriminator (MSD), were introduced to increase the generated speech's realism. While MSD works across numerous resolutions of the input audio to discover inconsistencies in different time-scales, MPD works on the periodicity of the audio by reshaping it into 2D representations. In order to stabilize training and enhance output quality, the authors also combine feature matching loss, mel-spectrogram loss, and adversarial loss. The end product is a system that performs noticeably better than its predecessors in terms of both synthesis speed and perceptual quality, producing 22.05 kHz audio with nearly human quality in real-time or at faster speeds.

The second study, "A Voice Cloning Method Based on the Improved HiFi-GAN Model," by Qiu et al., expands on the framework established by the original HiFi-GAN[8] model to accommodate voice cloning applications. The goal of voice cloning is to use a small number of voice samples to

create speech that mimics the unique vocal traits of a particular person. The ineffectiveness of autoregressive vocoders like WaveNet [7] and insufficient feature representation from speaker embeddings, particularly d-vectors, were the two main drawbacks the researchers found in the current systems. The study suggests a number of architectural enhancements to get around issues.They start by substituting the x-vector, which comes from a Time-Delay Neural Network (TDNN), for the d-vector. The x-vector improves the similarity of cloned voices by capturing more robust speaker features, particularly throughout complete utterances. Second, a Competitive Multiscale Convolution (CMSC) approach is used to process the input mel-spectrograms, which enables the model to extract more detailed local and global spectral information. Third, depthwise separable convolutions (DSC) are used in place of all conventional 1D convolutions to increase computational efficiency. This results in a considerable reduction of the model's parameter size of over 68% and an increase in inference speed of almost 31% on CPU.The combination of these changes leads to improved performance in both quality and speed, with increases observed in Mean Opinion Score (MOS) and Perceptual Evaluation of Speech Quality (PESQ) scores. The paper concludes that the enhanced HiFi-GAN[8] model is highly suitable for real-time voice cloning tasks, especially in applications demanding high speaker similarity and fast response times.

The third article, "Deepfake Audio Detection System,"[3] examines the similarly significant topic of detecting altered audio, whereas the first two papers concentrate on voice production. This paper tackles the developing issue of deepfake audio, which has consequences for digital identity theft, fraud, and disinformation. Mel Frequency Cepstral Coefficients (MFCCs) are the main feature extraction method used in the detection system that the researchers suggest, which is based on traditional machine learning methods. In the field of voice processing, MFCCs are highly acclaimed for their ability to compactly capture the spectral features of audio, which makes them perfect for identifying minute acoustic distinctions between synthetic and genuine speech.MFCC characteristics are extracted from audio samples by the suggested detection system and then fed into a number of machine learning classifiers, including as Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost [9]. The robustness and ensemble nature of Random Forest, which mixes the outputs of several decision trees to accomplish dependable categorization, made it stand out among these. The study divides deepfake audio into three categories: replay-based (reassembling real audio segments), imitation-based (using wavelet-based algorithms or human impersonation), and synthetic-based (text-to-speech systems as

Tacotron2 and HiFi-GAN). The authors show through experimental assessment that their system achieves a respectable detection accuracy, with robustness ensured by error message control and cross-validation.

There is a noticeable overlap between these publications and the current research, which is divided into two sections: [1] voice generation using HiFi-GAN and [2] fake voice detection using ML techniques. Both the original and updated HiFi-GAN papers are directly aligned with the initial section of this project. The pipeline outlined in both papers is mirrored in usage of Tacotron2 to extract mel-spectrograms and HiFi-GAN to convert them to audio. But in this project also has the ability to clone voices by using an audio sample as input and producing fresh speech in the same voice.The enhanced HiFi-GAN model, which emphasizes speaker-specific synthesis using x-vectors, is especially compatible with this method. The architecture and findings of Qiu et al. would be extremely pertinent to further refine our approach for quicker and more precise synthesis if our system makes use of speaker embeddings in a similar manner.

This project adds depth to the detection side by experimenting with several classification methods, such as SVM, KNN, and XGBoost, in addition to Random Forest. Compared to the method used in the "Deepfake Audio Detection System" publication, which mostly relies on conventional feature extraction and random forest models, this wider scope enables a comparative analysis of classifier performance and is more thorough. This application of feature engineering and contemporary machine learning approaches may increase detection accuracy, especially in the face of increasingly lifelike speech clones produced by models such as HiFi-GAN.

Each of the examined studies has limitations in addition to its strengths. Despite being revolutionary in terms of synthesis quality, HiFi-GAN ignores ethical concerns and abuse detection. The enhanced HiFi-GAN model is susceptible to abuse as it concentrates on technical improvement without incorporating any kind of authenticity verification. On the other hand, the detection study only offers fundamental machine learning techniques without investigating more sophisticated approaches such as adversarial training for robust generalization, spectrogram image

analysis, or deep learning (e.g., CNNs or RNNs). Furthermore, it only uses MFCC characteristics, which may not fully capture the subtleties of synthetic speech produced by contemporary TTS systems despite their strength.

To sum up, these articles together provide crucial insights into the two-pronged problem of producing and identifying deepfake audio. While the audio detection system gives fundamental methods for detecting modified audio using conventional ML and MFCC features, the HiFi-GAN family of models provides high-performance solutions for real-time, high-fidelity speech synthesis and voice cloning. our idea, by integrating these two realms, not only proves the usefulness of these state-of-the-art technologies but also pushes the envelope by merging synthesis and detection into one unified system. In the current environment, where synthetic media is becoming more and more capable and dangerous, this kind of integration is essential.

CHAPTER 3

OVERVIEW OF TECHNOLOGIES

3.1 Technologies Used

Python: This project is mostly built in Python, one of the world's most powerful and user-friendly programming languages. Python is perfect for creating deep learning applications because of its ease of use and readability as well as its extensive ecosystem of machine learning and audio processing modules. The whole voice cloning system workflow is implemented in this project using Python, including reading and preprocessing audio, creating deep learning models, making inferences, and managing I/O tasks like saving and playing audio files.

Guido van Rossum developed Python, a high-level, general-purpose programming language that was initially made available in 1991. Python is well-known for its easy-to-read syntax and support for a variety of programming paradigms, such as procedural, functional, and object-oriented. It has a large standard library, dynamic typing, and intelligent memory management. The Python Software Foundation maintains Python, which is open-source. Numerous operating systems have own interpreters. Python's versatility—it's utilized in scientific computing, data research, machine learning, web development, and more—is one of its greatest advantages. It is perfect for both novices and specialists due to its ease of use and robust libraries. Python serves as the project's core, enabling the smooth integration of processes for data processing, model creation, deep learning libraries, and audio production.

3.2 Libraries Used from Python

- **PyTorch:** PyTorch, an open-source machine learning library created by Facebook AI, is one of the project's main libraries. PyTorch is a popular tool for deep learning model training and deployment, and it offers dynamic computational graphs. PyTorch is used in this project to load and run the pretrained HiFi-GAN vocoder, which produces realistic audio, as well as to construct the custom generation network that predicts mel spectrograms from text and speech embeddings. Additionally, it handles model evaluation, applies operations, and manages tensors.

- **Torchaudio:** Torchaudio is a crucial library that enhances PyTorch by providing capabilities for audio data processing. Torchaudio may assist with simple audio operations and be helpful when

developing more complex audio pipelines, even though it is not extensively utilized in every aspect of the project. It functions as a component of the environment for the smooth handling of audio tensors in this project, which are built into the PyTorch framework.

- **Librosa:** One essential Python library made especially for audio and music analysis is called Librosa. In order to extract the speaker's vocal characteristics, the voice sample must first be loaded and resampled to a fixed sample rate (16 kHz) using Librosa. It has signal processing facilities as well, albeit only its most basic loading and sampling features are utilized in this context.

- **Numpy**: Additionally, we make use of NumPy, the core Python module for numerical computation. It is widely used in scientific computing and facilitates effective array operations. In this project, array operations for the final audio waveform and intermediate mel spectrogram outputs are handled using NumPy. Additionally, it entails transforming the audio data into a format that can be saved as a WAV file.

- **SciPy:** NumPy is the foundation for SciPy, another package for scientific computing. In this project, the created waveform is saved into a.wav file using the scipy.io.wavfile module, which allows it to be played again and reviewed. This stage preserves the output in a common audio format, completing the voice cloning process.

- **Matplotlib:** used in the final execution flow. However, it is often included in such projects to visualize spectrograms, waveform plots, or training curves, and it can be easily added to visualize the intermediate mel outputs.

- **Resemblyzer:** Resemblyzer is a speech encoder that uses deep learning. It transforms a speech signal into a fixed-length speaker embedding that records each individual's distinct vocal characteristics. In our project, this speaker embedding is extracted from an uploaded WAV file using Resemblyzer, and it is then sent to the generator network to affect the synthesized speech's voice.

- **IPython.display:** The produced audio is played directly in the Google Colab notebook using IPython.display. By leveraging Audio() objects to enable inline audio playback, it offers a convenient method of listening to results without having to download the file outside.

3.3 Frameworks Used

- **HiFi-GAN:** The HiFi-GAN (High-Fidelity GAN) framework is the most important one in this research. A cutting-edge vocoder model called HiFi-GAN uses mel spectrograms to produce high-quality audio waveforms. It learns to translate spectrograms—a representation of sound in time and frequency—into speech that sounds natural. The output of our GAN-based text-to-mel model is transformed into real audio in this project by loading a pretrained HiFi-GAN generator as the last stage. The output produced without HiFi-GAN would merely be a picture of sound, not real audio.

3.4 Platforms Used

- **Google Colab:** Users may write, run, and share Python code in a Jupyter Notebook environment with Google Colab, a free cloud-based platform created by Google. Because it provides free access to GPUs and TPUs, which expedites model training and testing, it is particularly helpful for data science and deep learning jobs. For researchers and students who might not have access to expensive technology, this makes it perfect. In our project, Google Colab makes it easier to integrate models like HiFi-GAN and Resemblyzer, execute code interactively, upload voice samples, and enjoy the produced audio—all within the browser. With its smooth file management via Google Drive and compatibility for several pre-installed libraries, it's a strong, scalable tool for effectively carrying out machine learning operations.

CHAPTER 4

IMPLEMENTATION

4. IMPLEMENTATION

4.1 Extracting Speaker Voice Embedding

Uploading a voice sample of the target is the first step in the voice cloning process. The audio in this sample is first cleaned up by preprocessing, which includes deleting noise, modifying the sampling rate, and making sure the format is constant. Following preprocessing, the audio is cleaned and sent into the Resemblyzer library, which creates a unique numerical representation of the speaker's speech using a deep neural network. We call this a speaker embedding. Tone, pitch, speaking style, and accent are among the vocal characteristics it records. The model will thereafter be able to generate new speech that sounds like the same person, even for terms they have never uttered before, thanks to this embedding, which functions as a fingerprint for the speaker's voice.

4.2 Converting Text into Numerical Format

The next step is to transform the intended input text into a machine learning-compatible format after getting the speaker's voice embedding. Since computers cannot comprehend letters, we first establish a vocabulary that includes all of the characters we want to employ, including punctuation, lowercase alphabets, and a padding sign. This vocabulary is then used to map each character in the phrase to a matching number. A numerical tensor representing the full statement is the end result. Our Generator model uses this tensor as input so that it may "know" what should be included in the output voice.

4.3 Generating Mel-Spectrogram using a Custom Generator

At the core of the system is the Generator model, which generates a mel-spectrogram as the output by combining the identity (speaker embedding) and the content (text). An illustration of how various sound frequencies vary over time is called a mel-spectrogram. Our unique neural network has linear layers to translate internal outputs into 80-dimensional mel-spectrogram frames, LSTM layers to represent temporal patterns, and embedding layers for the text. To create this spectrogram, the model combines and analyzes the speaker embedding and text tensor. Although it is still not genuine audio, at this point we effectively have a "visual sound map" of the speech in the speaker's voice.

4.4 Converting Spectrogram to Audio using HiFi-GAN

HiFi-GAN, a high-fidelity vocoder that transforms spectrograms into actual audio waveforms, receives the mel-spectrogram when it is ready. Generative Adversarial Networks, or GANs, are the foundation of HiFi-GAN and are particularly adept at producing realistic results. A discriminator in HiFi-GAN makes sure the audio sounds natural and human-like, while the generator uses the spectrogram to create audio that fits the intended frequency patterns.

We save time and computation by using a pretrained HiFi-GAN model in our project instead of training from scratch. A complete audio waveform that sounds as though it was uttered by the original speaker is the end result of this procedure.

4.5 Saving and Playing the Cloned Voice Output

The resulting audio waveform must then be transformed into a playable.wav file and saved locally as the last step. We save the waveform in the common 22,050 Hz audio format using Python's scipy module. We save the file and then play it straight in the Colab notebook using IPython's built-in audio playback facilities. The input statement is now audible in the cloned voice of the individual whose sample we submitted. The voice cloning pipeline is now complete, having gone from text and voice samples to completely produced speech audio.

4.6 Dataset Collection and Preprocessing

The initial stage in implementing false speech identification was to categorize the dataset into two types: actual and fake audio samples. Within an organized directory, these audio samples were kept in

their appropriate folders. The librosa library was used to load each WAV file. The audio data was adjusted before to feature extraction in order to guarantee amplitude uniformity between samples, which enhances machine learning model performance. To preserve data integrity, preprocessing also involved looking for faulty files or unintelligible audio, which were omitted.

4.7 Feature Extraction

We used Mel Frequency Cepstral Coefficients (MFCC) and their first and second-order derivatives (Delta and Delta-Delta) to extract relevant characteristics from the audio in order to reliably differentiate real and false voices. To capture pitch-related and spectral distribution patterns, respectively, chroma characteristics and spectral contrast were also calculated in addition to MFCCs. A rich and normalized representation of the acoustic characteristics of each sample was obtained by averaging these features and combining them into a single vector per audio file. By taking this step, models were guaranteed to get reliable and instructive training and prediction input.

4.8 Model Training and Evaluation

We implemented and compared multiple machine learning models: XGBoost, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The dataset was split into training and testing subsets using stratified sampling to preserve class distribution. Each model was trained on the training set and evaluated on the test set to compute accuracy. To improve performance, hyperparameters like the number of estimators, depth, and kernel types were carefully tuned. After training, each model was saved using the “joblib” library, allowing for later use in real-time voice predictions.

4.9 Voice Prediction and Result Analysis

A custom prediction function was used to test the trained models on a new audio sample. This sample underwent the same preprocessing and feature extraction steps as the training data. Each trained model then classified the sample as either "Real Voice" or "Fake Voice." Additionally, the accuracy of each model was printed alongside its prediction to offer insight into both performance and reliability. A final comparison table displayed all models’ accuracies in descending order, clearly highlighting the most effective approach SVM achieving the highest accuracy in detecting fake voices.

CHAPTER 5

RESULTS

Results

The text-to-speech pipeline goes as follows:

1. Text preprocessing : First, the input text is encoded into a list of symbols. In this tutorial, we will use English characters and phonemes as the symbols.
2. Spectrogram generation : From the encoded text, a spectrogram is generated. We use the Tacotron2 model for this.
3. convert to waveform using HiFi GAN : A complete audio waveform that sounds as though it was uttered by the original speaker is the end result of this procedure.

Character-based encoding

In this section, we will go through how the character-based encoding works.

Since the pre-trained Tacotron2 model expects specific set of symbol tables, the same functionalities is available in torchaudio. However, we will first manually implement the encoding to aid in understanding.

First, we define the set of symbols '_-!\'(),.:;? abcdefghijklmnopqrstuvwxyz'. Then, we will map the each character of the input text into the index of the corresponding symbol in the table. Symbols that are not in the table are ignored.

```
text = "Hello world! Text to speech!"
print(text_to_sequence(text))
```

[19, 16, 23, 23, 26, 11, 34, 26, 29, 23, 15, 2, 11, 31, 16, 35, 31, 11, 31, 26, 11, 30, 27, 16, 16, 14, 19, 2]

Phoneme-based encoding is similar to character-based encoding, but it uses a symbol table based on phonemes.

Similar to the case of character-based encoding, the encoding process is expected to match what a pretrained Tacotron2 model is trained on. `torchaudio` has an interface to create the process.

Spectrogram Generation

It is easy to instantiate a Tacotron2 model with pretrained weights, however, note that the input to Tacotron2 models need to be processed by the matching text processor.

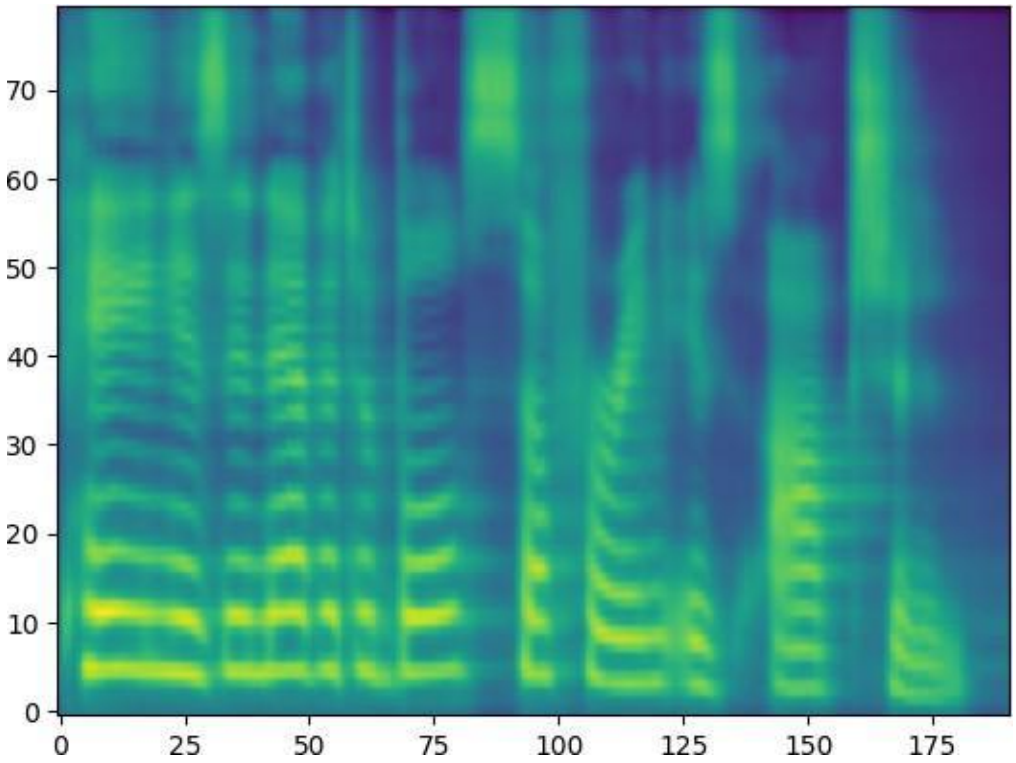


Fig. 5.1 Mel-Spectrogram of the given text

Mel-Spectrogram to Waveform Conversion (HiFi-GAN)

We employed HiFi-GAN, a generative adversarial network optimized for high-quality, real-time audio synthesis, to create audio from the mel-spectrogram.

- Generator:
  - Mel-spectrogram is the input that the generator uses.
  - Transposed convolutions are used to upsample it, and residual blocks (MRF module) are used for processing.
  - A raw waveform (float32 array) is output.

- Discriminators (used only during training):
    - Multi-Scale Discriminator (MSD) : The Multi-Scale Discriminator (MSD) assesses both the full and downsampled audio.
    - Multi-Period Discriminator (MPD) : The Multi-Period Discriminator (MPD) analyzes the waveform's periodic structures.
    - Through adversarial loss during training, these discriminators assist the generator in learning realistic speech patterns.

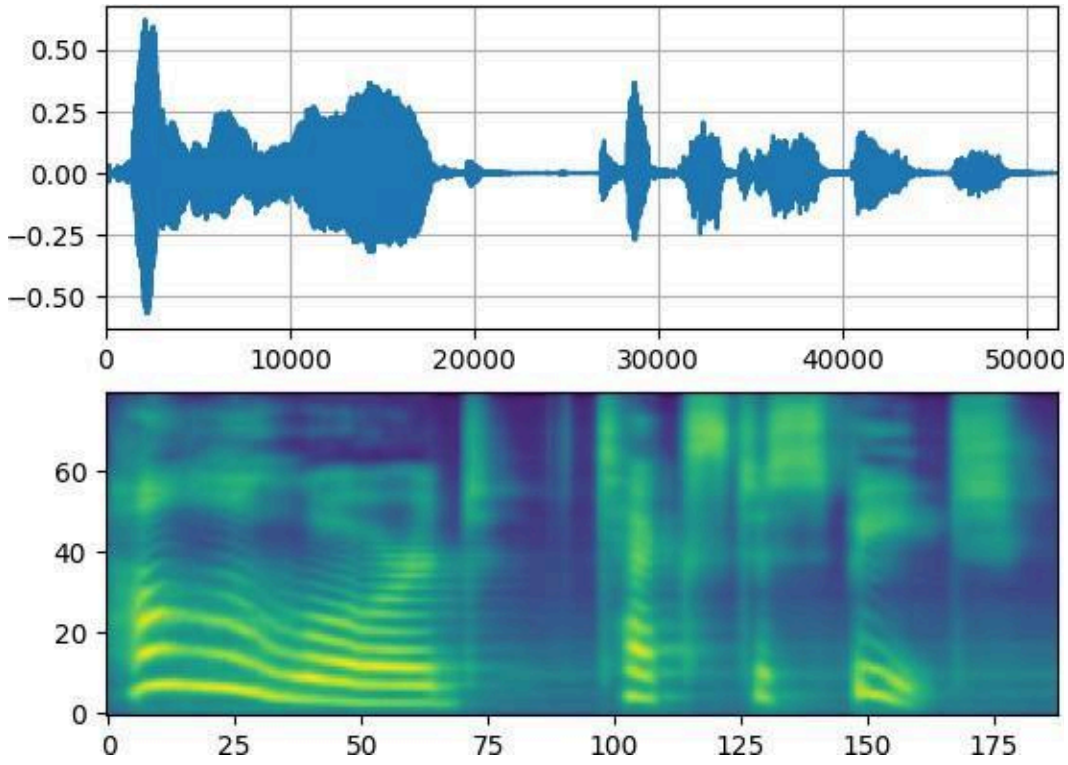


Fig. 5.2 WaveForm of the given text

While the existing system is highly accurate, there are a number of areas where the project may be expanded and improved. First, we intend to enhance the collection by integrating more diversified and high-quality false samples created by various speech synthesis techniques. This improves the model's generalization capabilities.

Although the present technique is effective for single-sentence cloning, further enhancements might include training the generator on a bigger dataset to enable broader generalization across diverse voices and accents. Adding emotion management or prosody variation could help in producing more expressive and natural-sounding speech.

Detection phase results

The implemented fake audio detection system was evaluated using multiple machine learning algorithms, including XGBoost, Random Forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). After extracting features such as MFCC, delta, chroma, and spectral contrast from real and fake audio samples, the models were trained and tested.

Features extracted:

1. MFCC (Mel-Frequency Ceptral Coefficients) : captures short term power spectrum of sound
2. Delta MFCC : captures how MFCC's change over time. useful for pitch changes and inflection in voices.
3. Delta-Delta MFCC : captures rate of change of delta MFCCs, useful to identify the finer time-based variations.
4. Chroma Features : Captures the energy distribution across the pitch classes. Used to show the harmonic difference (Like musical notes).
5. Spectral Contract : Captures peak and valleys in the Spectrum, helps in differentiating the timbre and richness of the voice.
6. Zero Crossing rate : Captures the noisiness present in the audio files.
7. RMS Energy : Captures the loudness and the variation present in the audio files.
8. Tonnetz : for Harmonic Pitch reactions.

Algorithms used:

1. **XGBoost** : Extreme Gradient Boosting, or XGBoost, is a potent and scalable machine learning method that has become well-known for its excellent results in regression and classification problems, particularly when applied to structured data. It is predicated on the idea of gradient boosting, an ensemble approach in which mistakes generated by earlier models are corrected by successively adding new models. In XGBoost, decision trees are constructed sequentially, with each new tree maximizing a given loss function in an attempt to reduce the residual errors of the aggregated prior trees. XGBoost is distinguished from other boosting algorithms by its superior tree pruning, regularization, and speed. It assesses a split's quality using second-order derivatives (Hessian matrix), leading to quicker and more precise training. In order to avoid overfitting, it further incorporates L1 and L2 regularization terms, which is especially helpful when working with high-dimensional audio characteristics such as MFCCs.

   XGBoost may greatly improve the accuracy of classifying genuine and synthetic audio in the context of this project, which entails recognizing phony speech produced by TTS systems like HiFi-GAN. Key frequency-based properties that may be given into the XGBoost classifier are represented by MFCC features, which are derived from audio signals. By examining these characteristics across a large number of training samples, the algorithm discovers trends and differences between real and fake speech. For example, unexpected frequency transitions or periodic abnormalities that are modest but persistent across samples may be present in cloned voices. By constructing several decision trees that concentrate on distinct facets of the feature space and progressively enhancing the prediction through weighted aggregation, XGBoost is exceptional at spotting such intricate patterns.First, a structured dataset is created by extracting MFCCs from both actual and synthetic speech samples. After that, this dataset is divided into testing and training sets. In order to minimize classification error, XGBoost analyzes which features—that is, certain cepstral coefficients—contribute more to class distinction during training and modifies tree topologies appropriately. After training, the model may be used to determine, from its MFCC signature, if a new voice input is authentic or not.

A robust and interpretable classifier that manages non-linear relationships, prevents overfitting through regularization, and effectively processes large datasets is thus provided by incorporating XGBoost into this voice detection pipeline. These attributes are essential for addressing the complex task of deepfake audio detection.

2. **Random forest :** A popular ensemble learning method called Random Forest creates several decision trees and aggregates their results to provide reliable and accurate predictions. It works using the bagging (bootstrap aggregating) approach, which creates several subsets by randomly sampling the training dataset (with replacement). In the prediction phase, the final output is decided by majority vote (for classification tasks) or averaging (for regression tasks) after a decision tree has been trained individually for each subgroup. Random Forest is more effective than individual decision trees because it may lower variance without raising bias, which makes it less likely to overfit.In order to ensure that the trees are de-correlated and, therefore, more dependable when merged, it also adds additional randomization by choosing a random subset of characteristics at each node during splitting. Random Forest is renowned for its ability to manage high-dimensional data efficiently and continue to perform well even when a sizable amount of the input is missing or noisy.

   Random Forest is essential to classification in research, which focuses on identifying phony speech samples produced by synthetic voice models like HiFi-GAN. These characteristics are used to train the Random Forest model after the audio has been preprocessed and transformed into numerical representations using Mel-Frequency Cepstral Coefficients (MFCCs). The fundamental spectral characteristics of speech that frequently change little but considerably between synthetic and real voices are captured by MFCCs. By building several decision trees, each of which makes predictions based on various feature combinations, the Random Forest model is able to understand these differences. The model determines which MFCC elements are most important for distinguishing between natural speech patterns and those produced by artificial generation methods during training.

   The first step in the detection process is data collecting, which involves gathering both real and fraudulent speech samples. MFCC, which converts raw audio into a fixed-size matrix of

frequency coefficients, is then used for feature extraction. After that, this matrix is identified as true or false and flattened into feature vectors. The Random Forest technique uses these labeled vectors as input to create many decision trees depending on various data subsets. The same feature extraction procedure is applied to a fresh, unseen speech sample, and the trained Random Forest model is given the MFCC vector. The class with the highest votes is selected as the final output, indicating whether the voice is synthetic or natural. Each decision tree in the forest makes a prediction.

3. **SVM :** A popular supervised machine learning approach for classification problems is Support Vector Machine (SVM), which can handle high-dimensional data and produce extremely accurate models. Finding the appropriate hyperplane to divide data points of several classes in a feature space is the fundamental concept of support vector machines (SVM). This hyperplane optimizes the margin, or the separation between the closest points of each class (often referred to as support vectors) and the hyperplane itself, when dealing with linearly separable data. SVM maps the input characteristics into a higher-dimensional space where separation is feasible using kernel functions (such the Radial Basis Function or polynomial kernels) for data that isn't linearly separable.As a result, complicated, non-linear correlations between features can be captured using SVM. It works especially well in binary classification issues like real vs. false speech recognition because of its accuracy, stability, and good generalization to unseen data.

   SVM is utilized in this research to determine if an audio sample is authentic or a fake. Mel-Frequency Cepstral Coefficients (MFCCs), which describe the short-term power spectrum of sound and are very good at capturing the subtleties of vocal features, are extracted from voice recordings after they have been gathered and preprocessed. The feature vectors that are fed into the SVM model are formed by these MFCCs. The SVM gains the ability to discriminate between the distribution of MFCC characteristics for real speech samples and those generated by artificial TTS systems such as HiFi-GAN during training. In order to minimize misclassification, the model finds the best hyperplane that divides these two classes by the greatest amount.

   When a fresh speech sample is entered during detection, it is run through the learned SVM model after undergoing MFCC feature extraction. The classifier then classifies the new point as real or false based on which side of the hyperplane it resides on. SVM's capacity to function well

even in situations with sparse data or tightly overlapping classes a frequent problem when separating incredibly lifelike cloned voices from authentic ones is one of its main benefits for project. SVM is a useful model for guaranteeing the dependability and accuracy of fake speech detection system because of its versatility in using kernel techniques and accuracy in defining precise decision boundaries.

4. **KNN :** A straightforward yet powerful machine learning approach for classification and regression problems is K-Nearest Neighbors (KNN). Since it is an instance-based learning technique, no explicit model is constructed during training. Rather, it commits the training data to memory and bases its choices on the majority class of the "K" data points that are the closest to a particular input sample. Usually, a distance metric like the Euclidean distance is used to assess the similarity, however depending on the data structure, alternative metrics like the Manhattan or cosine distance may also be employed.The selection of "K" is important since a bigger K may smooth out differences, while a smaller K makes the model more susceptible to noise. Because KNN is non-parametric, it can adjust to a broad range of data distributions, which makes it very helpful in applications with complicated or irregular underlying data patterns. Using MFCC characteristics taken from audio files, KNN is employed in this fake voice detection project to categorize speech samples. Each voice sample is represented as a vector in a high-dimensional feature space following preprocessing and feature extraction.All labeled vectors—real or fake—are saved throughout training. The method calculates the distance between this test vector and every other training vector when a fresh, unlabeled speech sample has to be categorized. The class that appears most frequently among the top K nearest neighbors is then assigned. Because the result is immediately impacted by the surrounding samples in the dataset, KNN is hence extremely straightforward and interpretable.

   KNN helps this detection system by serving as an easy-to-implement and fine-tune baseline classifier. When there is a distinct grouping of data, such variations in spectral patterns or temporal cues that frequently distinguish produced speech from human speech, it works very effectively.For instance, KNN will successfully capture the difference between cloned sounds that deviate owing to prosodic irregularities or generation errors and actual voices that tend to cluster around specific MFCC ranges. Additionally, KNN is very flexible and offers outstanding

classification performance when paired with well-extracted features like MFCCs since it does not presume any previous model or data distribution. It is an excellent addition to other classifiers in ensemble detection pipeline due to its competitive accuracy, transparency, and ease of use.

Among all the classifiers, the SVM model achieved the highest accuracy of **96.99%**, followed by XGBoost with **88.88%** accuracy. The Random forest and KNN models performed comparatively lower, with accuracies of **84.722%** and **71.759%** respectively. A sample prediction was also made using the trained XGBoost model, which correctly identified the voice as fake. These results demonstrate that ensemble-based models like SVM and XGBoost are more effective for detecting fake audio compared to traditional classifiers. The experiment validates the importance of using robust models and diverse features for improving the reliability of fake audio detection systems.

```
=== Model Accuracy Comparison ===
           Model  Accuracy (%)
0            SVM     96.990741
1        XGBoost     88.888889
2  Random Forest     84.722222
3            KNN     71.759259

Total execution time: 1839.32 seconds.
```

Fig 5.3 Results ScreenShort

CHAPTER 6

CONCLUSION

6. CONCLUSION & FUTURE WORK

6.1. Conclusion

A GAN-based voice cloning system that transforms provided text into realistic audio in the target speaker's voice was successfully built in this project. We constructed a comprehensive voice synthesis pipeline by integrating speaker embeddings (from Resemblyzer), a customized mel-spectrogram generator, and a HiFi-GAN vocoder for waveform production.

We used machine learning techniques to effectively create a false audio detection system. By collecting significant audio characteristics including MFCCs, chroma, and spectral contrast, the system successfully differentiates between authentic and fraudulent speech samples. We used a labeled dataset of produced and actual audio files to train and assess four distinct models: XGBoost, Random Forest, SVM, and KNN. The XGBoost classifier outperformed the others in identifying false voices, achieving the greatest accuracy.

6.2 Future Work

While the existing system is highly accurate, there are a number of areas where the project may be expanded and improved. First, we intend to enhance the collection by integrating more diversified and high-quality false samples created by various speech synthesis techniques. This improves the model's generalization capabilities.

Although the present technique is effective for single-sentence cloning, further enhancements might include training the generator on a bigger dataset to enable broader generalization across diverse voices and accents. Adding emotion management or prosody variation could help in producing more expressive and natural-sounding speech.

CHAPTER 7

REFERENCES

7. REFERENCES

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